Design and Characterization of Low resolution spectrometer-

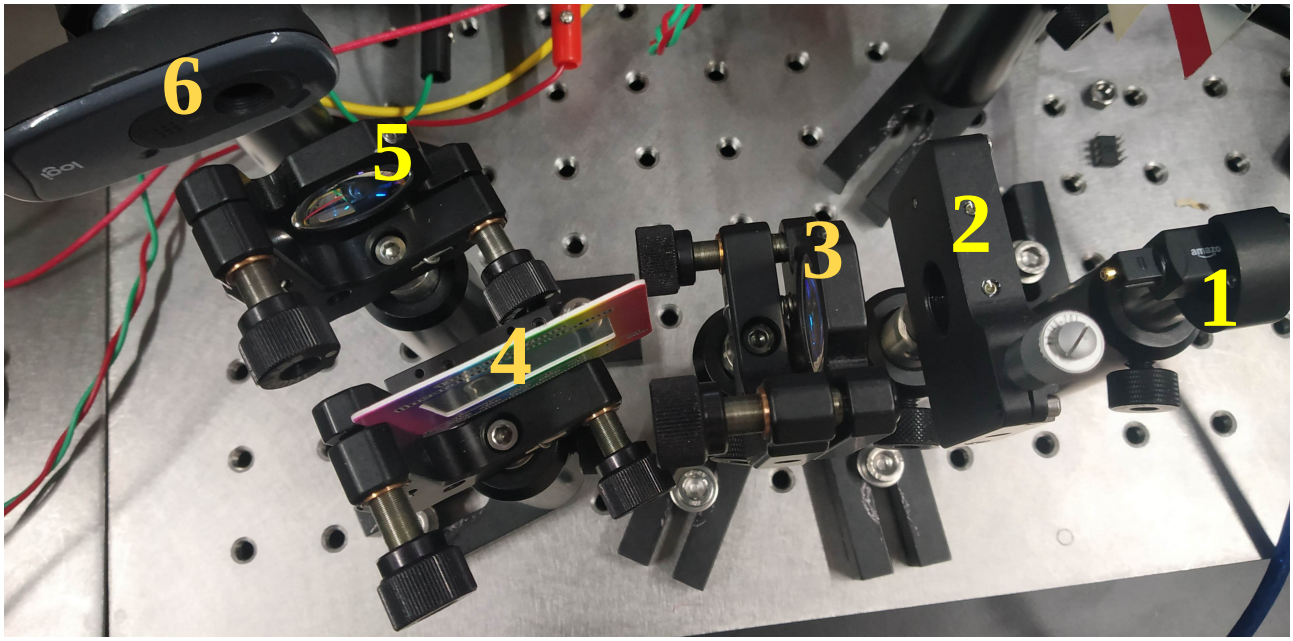

Fig. 1. Current configuration of low resolution spectrometer.

Components-

1. fibre cable with 1mm aperture, 2. variable slit, 3. convex lens (30 mm),

4. transmission grating (1000 l/mm), 5. convex lens (50 mm), 6. webcam (1280×960 pixel)

Calibration of spectrometer-

Initial calibration has been done using pulse dye laser (calibrated with iodine cell) and He – Ne laser in the 588 nm – 632.8 nm range.

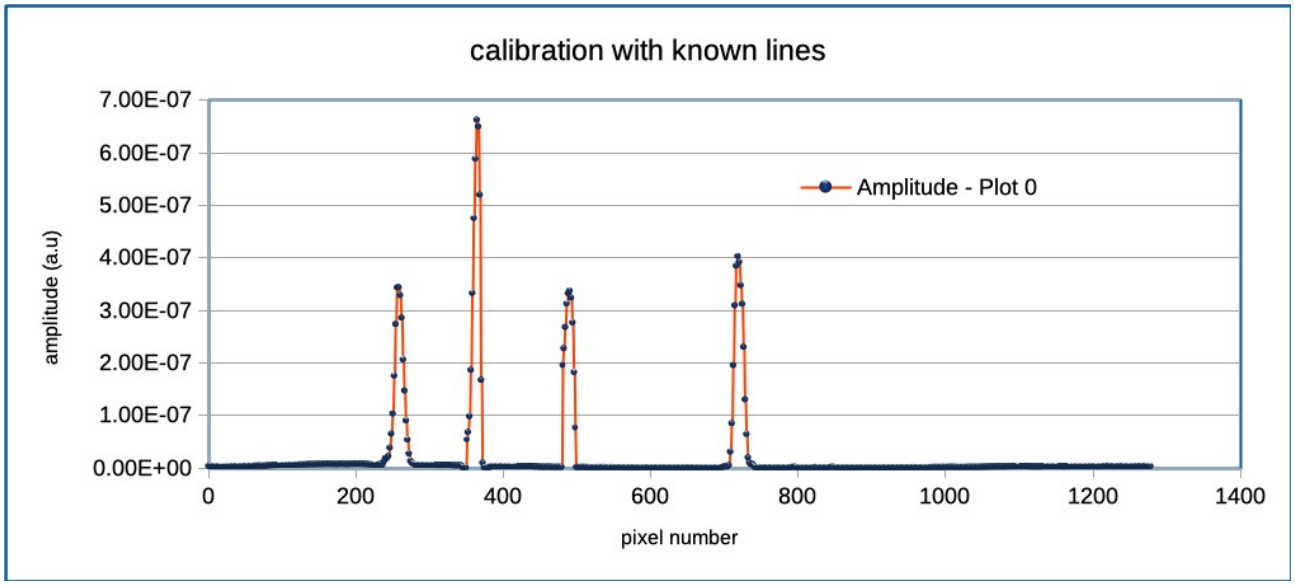

Fig.2 shows intensity plot of different wavelengths with the pixel number.. Each peaks are correspond to 588.188, 598.188, 610.188 and 632.816 nm (left to right), respectively.

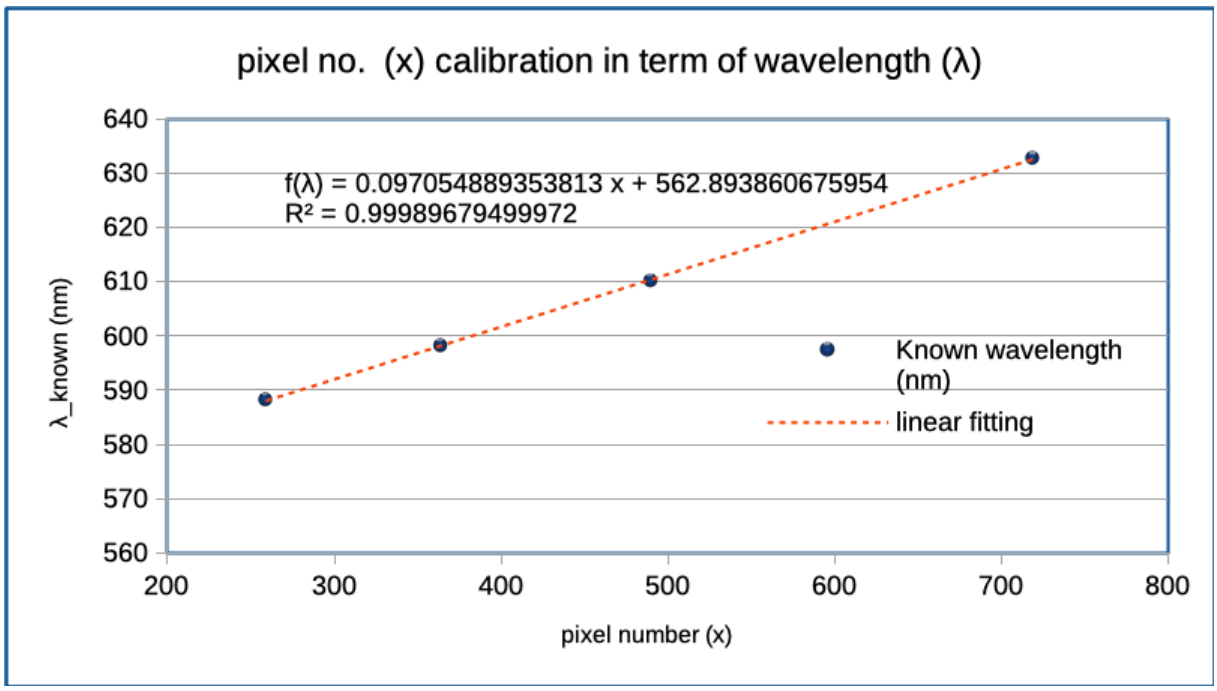

Fig. 3. known wavelength verses pixel number plot. Using the linear fit equation, we calibrated the pixel number to wavelength (nm).

Spectrum of Neon lamp-

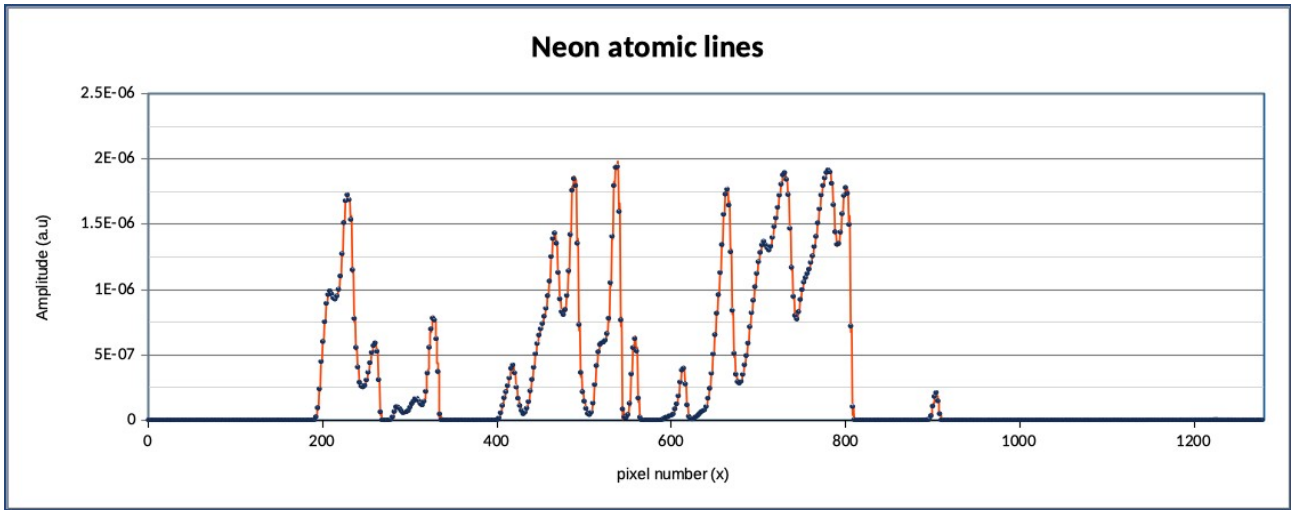

Fig. 4. (a) shows intensity plot of Neon atomic lines with pixel number**, (b)** 2D image of spectrum of Neon lamp.

- Neon atomic lines plotted as a function of wavelength using same fitted equation.

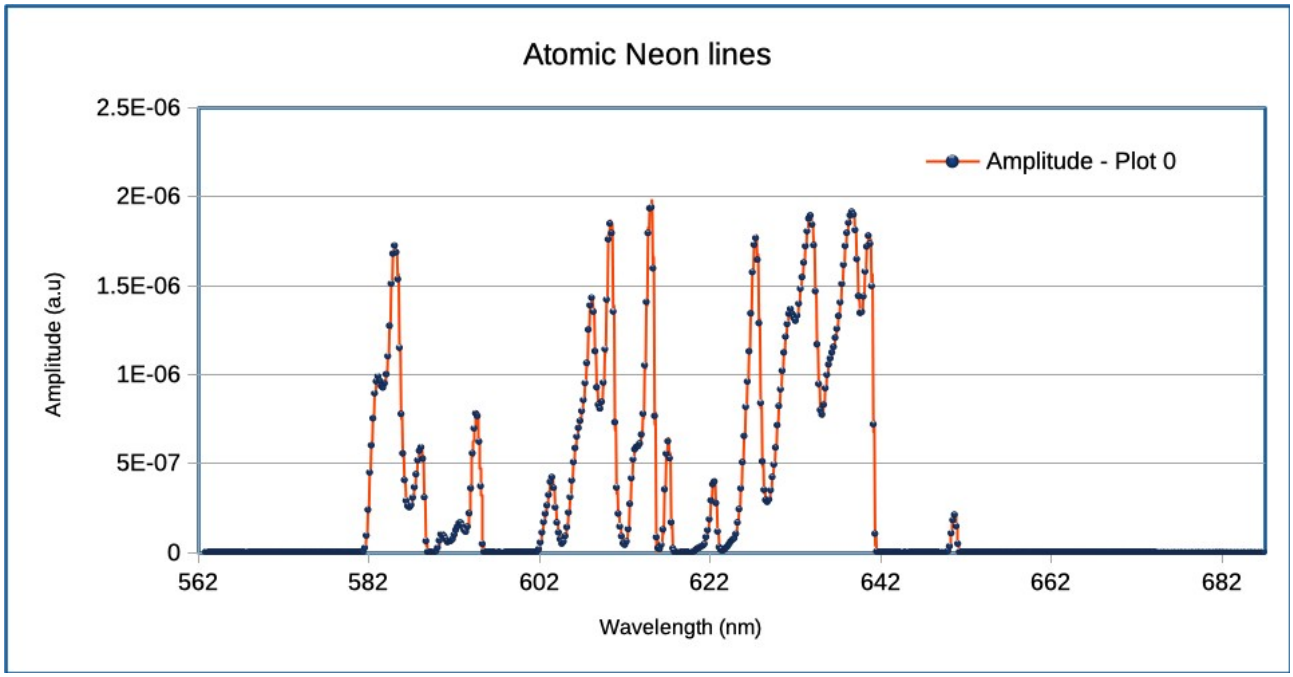

Fig.5 shows intensity plot of Neon atomic lines with calibrated wavelengths.

Table.1 In column 1 the calculated wavelengths are listed, column 2 shows known wavelength, and column 3 tabulate the difference values (nm) of calculated and reference wavelength of Neon lines.

Calculated wavelength λ_{cal}	**Known wavelength λ_{known}**	**$\lambda_{cal} - \lambda_{known}$ (nm)**
585.227	585.248	-0.021
588.208	588.189	0.019
590.852	590.642	0.21
594.774	594.483	0.291
597.957	597.562	0.395
603.591	602.992	0.401
608.262	607.433	0.829
610.439	609.616	0.82
615.152	614.306	0.846
617.194	616.359	0.835
622.562	621.728	0.834
627.448	626.649	0.799
631.982	632.816	-0.834
633.874	633.442	0.432
638.749	638.299	0.45
640.701	640.224	0.477
650.766	650.652	0.114

Conclusion-

- The accuracy of spectrometer was found ± 0.8 nm.
- The spatial resolution of spectrometer is typically 0.1 nm.
- The spectrometer can cover approximately 127 nm in full resolution mode (1280 × 960) of the webcam.